Two exact identities for the magnetic energy, a ground-state sum rule, and why pointwise Gibbs domination fails as Agmon does

The magnetic energy of the Kogut–Susskind Hamiltonian satisfies two exact relations on the configuration space $G^{\mathcal{E}}$:

$$\Delta V = 4C_2(N|\mathcal{P}| - V), \qquad |\nabla V|^2 \;\le\; 16\,V \quad (SU(2)),$$

the first because $V - N|\mathcal{P}|$ is an eigenfunction of the Laplace–Beltrami operator with eigenvalue $-4C_2$, the second from the completeness relation together with $\sin^2\theta = (1-\cos\theta)(1+\cos\theta)$. They give an exact sum rule for the ground state,

$$A'\int (V - N|\mathcal{P}|)\,|\nabla\Omega|^2 d\mu + A'(2C_2 - K)\langle V - N|\mathcal{P}|\rangle + B'\,\mathrm{Var}_\Omega(V) = 0, \qquad K = \int |\nabla\Omega|^2,$$

and give a sufficient positivity region for the comparison function $\varphi = e^{-\lambda V/2}$. Its threshold and prefactor depend on total ground energy and global normalization data. This supplies no volume-uniform local bound. The limitation concerns these particular comparison functions; it is not an impossibility theorem for Gibbs domination or every use of the eigenvalue equation.

Fix a finite periodic three-dimensional spatial cubic lattice with four distinct links per plaquette. Use normalized product Haar measure, the product bi-invariant metric with $\mathrm{tr}(T^aT^b) = \delta^{ab}/2$, and $V = \sum_p (N - \mathrm{Re}\,\mathrm{tr}\,U_p)$. Set $A' = g^2/2$, $B' = 2/g^2$ and $H = (\hbar c/a)[A'(-\Delta) + B'V]$. The smooth positive ground state Ω is normalized by $\int \Omega^2 d\mu = 1$ and satisfies $[A'(-\Delta) + B'V]\Omega = e_0\Omega$; its finite-lattice existence and simplicity are T1. The Laplacian identity extends to other four-distinct-link geometries; the numerical gradient estimate below uses the stated spatial incidence.

1. The Laplacian of the magnetic energy

Proposition 1. On $\mathcal{M} = G^{\mathcal{E}}$ with the product bi-invariant metric,

$$\Delta V = 4C_2(R_{\mathrm{f}})(N|\mathcal{P}| - V),$$

where $C_2(R_{\mathrm{f}})$ is the quadratic Casimir of the defining representation. Equivalently $W = V - N|\mathcal{P}|$ satisfies $\Delta W = -4C_2 W$: the magnetic energy, shifted by its Haar mean, is an eigenfunction of the Laplacian.

Proof. Fix a plaquette p and one of its four links ℓ. Writing $\mathrm{tr}\,U_p = \sum_{ab}(U_\ell)_{ab}M_{ba}$ with M the ordered product of the other three links, each matrix element of U_ℓ in the defining representation is an eigenfunction of Δ_ℓ with eigenvalue $-C_2(R_{\mathrm{f}})$, by Peter–Weyl. Hence $\Delta_\ell\,\mathrm{tr}\,U_p = -C_2\,\mathrm{tr}\,U_p$, and summing over the four links of p, $\Delta\,\mathrm{Re}\,\mathrm{tr}\,U_p = -4C_2\,\mathrm{Re}\,\mathrm{tr}\,U_p$. Summing over plaquettes, $\Delta V = \Delta\sum_p(N - \mathrm{Re}\,\mathrm{tr}\,U_p) = 4C_2\sum_p \mathrm{Re}\,\mathrm{tr}\,U_p = 4C_2(N|\mathcal{P}| - V)$. □

2. The gradient of the magnetic energy

Proposition 2. On the periodic three-dimensional spatial cubic lattice with four plaquettes per link, for $SU(2)$, $|\nabla V|^2 \le 16\,V$ pointwise on $\mathcal{M}$. For general $SU(N)$, $|\nabla V|^2 \le 16\,N\,V$.

Proof. For one link of one plaquette, the completeness relation of the upper-bound note Corollary 2 gives

$$\sum_a |X_a^\ell\,\mathrm{tr}\,U_p|^2 = \tfrac{1}{2}\Big[\,\mathrm{tr}(MM^\dagger) - \tfrac{1}{N}|\,\mathrm{tr}\,M|^2\Big] = \tfrac{1}{2}\Big[N - \tfrac{1}{N}|\,\mathrm{tr}\,U_p|^2\Big].$$

For $SU(2)$, writing $\mathrm{tr}\,U_p = 2\cos\theta$, this is $1 - \cos^2\theta = \sin^2\theta = (1-\cos\theta)(1+\cos\theta)$, while $V_p = 2 - 2\cos\theta$, so

$$|\nabla_\ell\,\mathrm{Re}\,\mathrm{tr}\,U_p|^2 \le \sin^2\theta = \frac{V_p}{2}\Big(2 - \frac{V_p}{2}\Big) \le V_p.$$

Each link lies in four plaquettes, so by Cauchy–Schwarz $|\nabla_\ell V|^2 \le 4\sum_{p\ni\ell} V_p$, and summing over links, with four links per plaquette, $|\nabla V|^2 \le 4\sum_\ell \sum_{p\ni\ell} V_p = 16\sum_p V_p = 16V$. The general case replaces the bound $\sin^2\theta \le V_p$ by $\frac{1}{2}[N - |\operatorname{tr} U_p|^2/N] \le NV_p$, which follows from $|\operatorname{tr} U_p| \ge N - V_p$ when $V_p \le N$ and from the trivial bound otherwise. □

Both identities vanish at $U_p = 1$: a configuration with no magnetic energy has no magnetic gradient, as it must.

3. An exact ground-state sum rule

Proposition 3. With $W = V - N|\mathcal{P}|$, $K = \int |\nabla\Omega|^2 d\mu$, $\langle\cdot\rangle = \langle\Omega, \cdot\,\Omega\rangle$ and $\operatorname{Var}_\Omega(V) = \langle V^2\rangle - \langle V\rangle^2$,

$$A'\int W|\nabla\Omega|^2 d\mu + A'(2C_2 - K)\langle W\rangle + B'\operatorname{Var}_\Omega(V) = 0.$$

Proof. Multiply $-A'\Delta\Omega + (B'V - e_0)\Omega = 0$ by $W\Omega$ and integrate. Since

$$\int W\Omega\,\Delta\Omega = -\int W|\nabla\Omega|^2 - \int \Omega\,\nabla W\cdot\nabla\Omega,$$

$$\int \Omega\,\nabla W\cdot\nabla\Omega = -\tfrac{1}{2}\int (\Delta W)\Omega^2 = 2C_2\langle W\rangle,$$

by Proposition 1, so $\int W\Omega\Delta\Omega = -\int W|\nabla\Omega|^2 - 2C_2\langle W\rangle$ and the equation becomes $A'\int W|\nabla\Omega|^2 + 2A'C_2\langle W\rangle + B'\langle VW\rangle - e_0\langle W\rangle = 0$. Now $e_0 = A'K + B'\langle V\rangle = A'K + B'(\langle W\rangle + N|\mathcal{P}|)$ and $\langle VW\rangle = \langle W^2\rangle + N|\mathcal{P}|\langle W\rangle$, so the last two terms combine into $-A'K\langle W\rangle - B'\langle W\rangle^2 + B'\langle W^2\rangle$. □

The sum rule ties the magnetic variance of the ground state to its kinetic energy and to the correlation between magnetic energy and kinetic density. At $g \to \infty$ every term vanishes with B', and at finite coupling it is one exact constraint on the ground-state measure.

4. Pointwise Gibbs domination and where it stops

Try $\varphi = e^{-\lambda V/2}$, $\lambda > 0$, as a comparison function for $L = -A'\Delta + (B'V - e_0)$. Using $\Delta\varphi = \varphi[\frac{\lambda^2}{4}|\nabla V|^2 - \frac{\lambda}{2}\Delta V]$ and Propositions 1–2,

$$\frac{L\varphi}{\varphi} = B'V - e_0 - A'\frac{\lambda^2}{4}|\nabla V|^2 + A'\frac{\lambda}{2}\Delta V \;\ge\; \Big[B' - 4A'\lambda^2 - 2A'\lambda C_2\Big]V + \Big[2A'\lambda C_2 N|\mathcal{P}| - e_0\Big],$$

for $SU(2)$, using $|\nabla V|^2 \le 16V$ and Proposition 1.

Proposition 4. If λ satisfies $B' > 4A'\lambda^2 + 2A'\lambda C_2$, then $L\varphi \ge 0$ on $\{V \ge V_*\}$ with

$$V_* = \frac{(e_0 - 2A'\lambda C_2 N|\mathcal{P}|)_+}{B' - 4A'\lambda^2 - 2A'\lambda C_2},$$

and the maximum principle gives

$$\Omega(U) \;\le\; \Big(\max_{\{V\le V_*\}}\frac{\Omega}{\varphi}\Big)\, e^{-\lambda V(U)/2} \qquad \text{for all } U.$$

Proof. Set $u = \Omega/\varphi$. From $L\Omega = 0$, $0 = -A'\varphi\Delta u - 2A'\nabla u\cdot\nabla\varphi + u\,L\varphi$. At an interior maximum of u one has $\nabla u = 0$ and $\Delta u \le 0$, so the first two terms are ≥ 0 and $u\,L\varphi \le 0$; since $u > 0$, the maximum lies in $\{L\varphi \le 0\} \subset \{V \le V_*\}$. □

Where it stops. With $e_0 \le B'N|\mathcal{P}|$, the threshold obeys

$$V_* \;\le\; N|\mathcal{P}|\,\frac{B' - 2A'\lambda C_2}{B' - 4A'\lambda^2 - 2A'\lambda C_2} \;\xrightarrow[\lambda\to 0]{}\; N|\mathcal{P}|,$$

This is an upper bound on the threshold, not a proof that its actual value is asymptotic to the Haar mean. The resulting comparison uses a constant determined on the global sublevel $\{V \leq V_*\}$. A local event need not force a global excess. No volume-uniform local probability estimate follows from the available bounds on that threshold and normalization constant, as in the Agmon argument.

5. Global comparisons and local probability

The two comparisons use the eigenvalue equation $-A'\Delta\Omega+(B'V-e_0)\Omega = 0$ and its total ground energy. The weighted identity and exponential trial function control global potential excess. Their failure to prove local suppression does not exclude a different Hamiltonian argument.

The Wilson density $e^{-S_w} = \prod_p e^{-(2/g_E^2)V_p}$ prices an action change locally, but shared links and the partition function prevent this unnormalized factorization from being a probability theorem. The slice representation is a different exact identity at fixed temporal spacing. Passing to the Hamiltonian requires convergence of the ground-state projectors; transferring a local bound requires the same anisotropic slice observable and constants uniform in all claimed limits.

6. Consequence for STATE

The exact Laplacian/gradient identities and sum rule survive, with the setting above and the periodic three-dimensional plaquette incidence used in the gradient proof. Proposition 4 gives the stated global comparison; it neither proves nor disproves every possible Gibbs domination bound. The Agmon tail also controls a global excess with finite-volume constants.

No normalized interacting local suppression $Ce^{-c\eta^2/g^2}$ follows from either eigenfunction comparison, plaquette action factorization or the slice identity alone. The missing local probability and signed operator estimates remain separate.